



Technical Notes

An FFT-based beam profile denoising method for beam profile distortion correction[☆]Yinjie Lin, Ping Tan^{*}, Dong Li, Xinze Wang, Xinyi Shen

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ABSTRACT

The accurate verification of beam parameters is mandatory for treatment safety in particle therapy, therefore, the systematic uncertainties of the segmented ionization chambers and detector arrays used as beam monitors, must be measured. In this study, a Fast Fourier Transform (FFT) based denoising method is proposed to remove noise from the pencil beam profile signal in both the spatial and spatial-frequency domains. In the spatial domain, the bias caused by noise was corrected, and the noise outside the 4σ range of the spot center was discarded. Then, in the spatial-frequency domain, thanks to the Gaussian shape of the signal's magnitude spectrum, the 4σ range of frequency components was extracted to eliminate the signal fluctuation caused by noise. In addition, low amplitude high-frequency components are finely removed by thresholding to reduce the distortion of beam profile signals. The performance of the proposed method was verified through simulation experiments in terms of the coefficient of determination (R^2) and signal-to-noise ratio (SNR). Results show that SNRs of the denoised signals are more than six times higher than those of the original signals and R^2 of the denoised signals is larger than 0.9. The denoising capability of the presented method was verified by an X-ray radiation experiment. R^2 of the denoised signal achieves 0.96 from 0.60, and the fitting errors of beam parameters (μ and σ) also decrease by denoising without increasing the radiation intensity and the integration time.

1. Introduction

In pencil-beam scanning (PBS) system for proton therapy facility, it is mandatory to make solid determination of beam profile parameters during a treatment or quality assurance (QA). The beam position precision required at the isocenter should be within 1 mm [1–3]. The two-dimensional (2D) detectors, such as segmented ionization chambers (ICs) placed at the exit of nozzles, are used to monitor the beam fluence and the transverse beam profile for safety and to guide the treatment progress [4–7]. 2D arrays of hundreds of ionization chambers or diodes, generally placed at the isocenter plane are also used for beam characterization during the system commissioning and quality assurance [8]. However, noise contributed by front-end electronics interferes with signals acquired from these detectors [9,10]. For the strip detectors used in PSI and Italian National Center of Oncological Hadron-therapy (CNAO), the width of strips are 1.72 mm and 1.55 mm [2], respectively, and the corresponding pitch are 2.22 mm and 1.65 mm [4]. In CNAO, the area of each conductive pixel of pixel detectors is 6.5 mm $\times$ 6.5 mm with a pitch of 6.6 mm. While in PSI, size of each pixel is 2.2 mm $\times$ 2.2 mm with a pitch of 2.5 mm for a new developed pixel detector based on Printed Circuit Board

technology [11]. The spatial resolution for above detectors is in the range of 1.65 – 6.6 mm. Though the signal-to-noise ratio (SNR) per pixel increases with the enlargement of pixel size, the large scale of pixels leads to too few pixels to sample the pencil beam profile and exacerbates the volume averaging of the Gaussian profile over the pixel sensitive volumes [12]. As the radiation intensity and the integration time is restricted by the accelerator performance as well as by the treatment plan requirements, the resulting SNRs of each detector channel may be insufficient to evaluate the beam profile accurately.

Researchers from Washington University applied digital smoothing filters to two-dimensional dose images that improved the dosimetric reliability and visualization of dose distributions [13]. In PSI, a 3×3 pixel median filter is applied to charged-coupled device (CCD) images for noise reduction [14]. Recently, a deep learning (DL) approach, the double U-Net model, was used to denoise the proton-induced secondary-electron-bremsstrahlung (SEB) images and produce higher-resolution dose images [15]. However, the spatial resolution of detector signals is much lower than that of images, that digital filters may not achieve satisfactory results, and deep learning approaches need databases for training and have high computation costs. Therefore, in this study, we

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focus on effectively denoising the detector data, especially the distorted data, for reliable beam profile verification. A Fast Fourier Transform (FFT) based beam profile denoising method was proposed to remove noise from the seriously noisy signal acquired from detectors while preserving the pencil beam information. The method adapts for both 1D and 2D detector data. It deals with the bias from the spatial domain and reduces signal fluctuation and distortion by removing high-frequency components. The FFT-based beam profile denoising method was tested under different SNRs in terms of coefficient of determination R^2 and SNR. In addition, it was verified using X-ray radiation experiments.

2. Beam and detector modeling

2.1. Simulation of the detector signals

This study is based on a simulation of the output from a pixel ionization chamber comprising 60×60 square pixels ($2 \text{ mm} \times 2 \text{ mm}$) [11]. Number of ion-electron pairs produced for ionization by the primary particles are generated using MATLAB R2020b following a 2D normal distribution with input parameters (μ_x , μ_y , σ_x , and σ_y), where μ_x and σ_x are the mean and standard deviation of the projected beam profile in the x -direction, respectively, μ_y and σ_y are defined similarly in the y -direction. In the study, $\sigma_x = \sigma_y = \sigma$. The signal value G_i of each pixel ($i \in [1, 3600]$) corresponds to the ionization charge and is proportional to the charges collected by each detector or electrode. An additional noise contribution D_i is added to the signal from each pixel according to a normal distribution with mean μ_N and sigma σ_N , where μ_N is a random number characterizing a certain bias, which follows a normal distribution with mean 0 and sigma σ_{μ_N} ($\sigma_{\mu_N} = 0.25\sigma_N$).

In the following, the signal-to-noise ratio (SNR), defined as the ratio of the average value of signals in a pixel to the standard deviation of noise per pixel, will be used to evaluate the strength of the signal compared to the channel noise:

$$SNR = \frac{\bar{G}}{\sigma_N} = \frac{\sum_{i=1}^W G_i}{W\sigma_N} \quad (1)$$

where $\bar{G}$ is the average value of G_i and W is the number of pixels in a detector.

Referring to the user manual of the commercial electrometer I128 (Pyramid Technical Consultants, Inc., Massachusetts, USA) connected to ionization chambers, the root mean square (RMS) of input noise charge σ_N at 1000 μs integration time, is less than 0.05 pC which corresponds to 3.12×10^5 protons [16]. In proton therapy, the beam intensity for treatment is typically 0.5–1 nA [17] and reaches the 4 nA at 230 MeV with the IBA Isochronous cyclotron [18]. The gain factor of the chamber with a gap of 5 mm filled with air, is typically 155 at 70 MeV and 65 at 230 MeV [19]. The time required to deliver one spot is between 1 and 10 ms, depending on the MU prescribed for the spot among other variables [20]. Besides, spots indicate fixed monoenergetic pencil beams. In this study, the FFT-based beam profile denoising algorithm is verified in the case of SNR between 0.2 and 1.5 as expected with clinical proton beams and as detector we considered a detector with 3600 pixels (W).

2.2. Spectrum analysis of the beam profile signal

Segmented electrodes in an ionization chamber act like a periodic pulse signal which uniformly samples the pencil beam profile signal in the x - or y -direction of the spatial domain.

In the case that the size of each pixel is $2 \text{ mm} \times 2 \text{ mm}$, the normalized beam signal in the cases of $\sigma=4 \text{ mm}$ and $\sigma=10 \text{ mm}$ and their corresponding discrete signals of the detector's 1D projection are shown in Fig. 1. As shown in Fig. 1, frequencies of pencil beam signals are concentrated in the low-frequency sub-band. As the spot size σ decreases, the amplitude of the low-frequency component decreases, and that of the high-frequency component increases. When $\sigma \geq 4 \text{ mm}$,

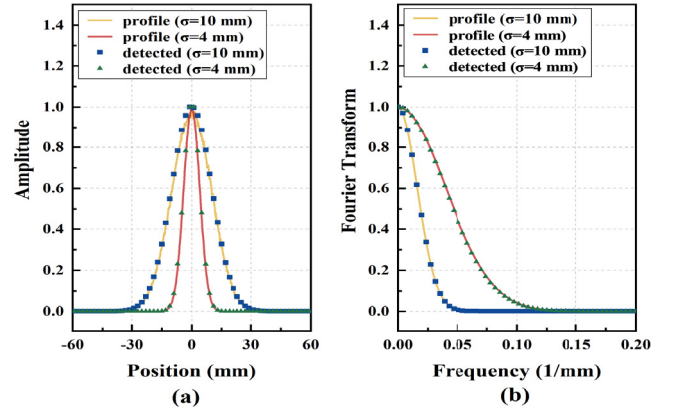


Fig. 1. Normalized pencil beam signal ($\sigma = 4 \text{ mm}$ and 10 mm) and their corresponding discrete signals of the detector's 1D projection (a), the corresponding Fourier transform (b).

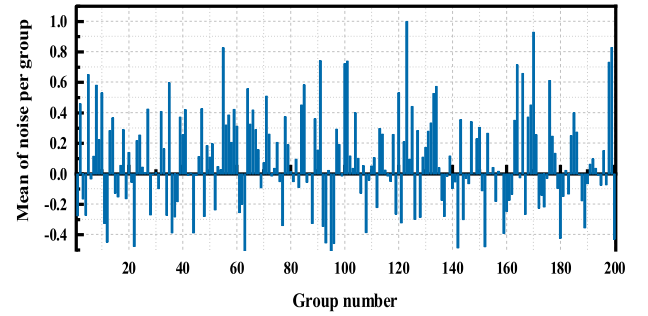


Fig. 2. Mean of noise data per group normalized to the maximum, in the case that the integration time is 100 μs , the number of groups is 200, and each group contains values of all channels of a detector for each integration time.

the highest spatial frequency of the pencil beam signal is less than 0.125/mm. Referring to Nyquist's sampling theorem, the distance between detectors in the 2D arrays should be less than 4 mm to attain the required 1D spatial sampling frequency.

2.3. Evaluation criteria

In the study, the coefficient of determination (R^2 or R-squared) is used to evaluate the difference between the noiseless pencil beam signal with the noised one, a value close to 1 means that there is little noise distorting the signal, and it is calculated as follows [21],

$$R^2 = 1 - \frac{\sum_{i=1}^W (\hat{G}_i - G_i)^2}{\sum_{i=1}^W (\bar{G} - G_i)^2} \quad (2)$$

where W is the number of pixels in a detector, G_i and $\hat{G}_i$ are the pixel noiseless and noised signals, respectively, and $\bar{G}$ is the mean of G_i .

3. The FFT-based beam profile denoising method

3.1. Denoising in the spatial domain

Compared with the white Gaussian noise, the mean of noise data per group measured using a commercial electrometer I128 connected to a multi-strip ionization chamber IC128 (Pyramid Technical Consultants, Inc., Massachusetts, USA) [22] is random both in sign and magnitude as shown in Fig. 2. Each group contains values of all channels of a detector for each integration time. Fig. 2 shows the mean of noise data per group normalized to the maximum, in the case that the integration time is 100 μs and the number of groups is 200. As an example, the noise amplitude

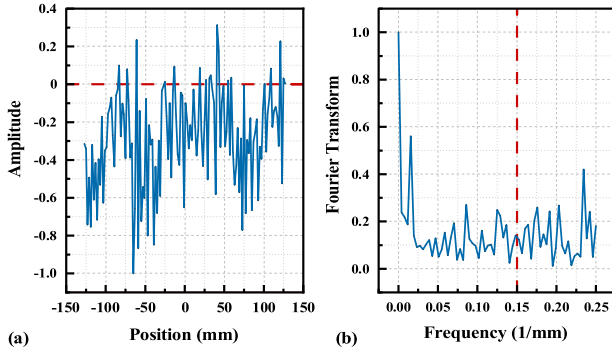


Fig. 3. Amplitude of noise in one group (a) and the corresponding Fourier transform (b).

and the Fourier Transform of one group are shown in Fig. 3. The noise with non-zero mean adds a bias to the detector signal and raises the uncertainty of the beam monitoring system. In addition, the noise with a non-zero mean has a zero-frequency component with high amplitude (Fig. 3(b)). However, the content of the pencil beam signal is mainly contained in the low-frequency portion. Therefore, the bias caused by noise cannot be effectively eliminated in the frequency domain. In the study, signals outside the 4σ range from the center of the pencil beam signal determined by Gaussian fitting are treated as noise. Therefore, the bias per group is evaluated as the mean of noise and eliminated by subtracting the estimated value from the detector signal.

In addition, signals within the 4σ range from the center of the detector signal are extracted, and those beyond this range are discarded as noises.

3.2. Denoising in the frequency domain

From the spectrum analysis of the detector signal during the beam delivery, we observed that the signal is mainly in the low-frequencies. Thus, the high-frequency components are reduced as follow:

I. From the frequency spectrum of the detector signal, extracted by FFT, zero frequency component is shifted to the spectrum center. The corresponding amplitude spectrum is shaped as a 2D Gaussian distribution with a standard deviation σ_{lf} determined by $\text{FWHM}/2.355$, where FWHM is full width at half maximum of a Gaussian distribution (Fig. 4(a)).

II. The high-frequency components beyond the $4\sigma_{lf}$ respect to the center of the spectrum are discarded from both the amplitude spectrum (Fig. 4(b)) and the frequency spectrum.

III. Remove the low amplitude components using the threshold $3\sigma_{hf}\sqrt{2 \cdot \log_{10}(N)}$ (Fig. 4(c)), where N is the number of the discarded elements, and σ_{hf} is their standard deviation.

IV. Discard the stray high-frequency components in the amplitude spectrum (Fig. 4(d)) and frequency spectrum.

V. Recover the detector signal by inverse discrete Fourier transform using the frequency spectrum.

Fig. 4 shows an example of denoising by the high-frequency components reduction in the case of $\sigma=6$ mm and SNR=0.2.

4. Validation of the FFT-based beam denoising method

Simulation experiments were designed to investigate the performances of the proposed denoising methods for different SNRs between 0.2 and 1.5. The number of simulations is 1000.

In the condition of SNR=0.2 and $\sigma=10$ mm, the original beam profile and denoised profiles for each step are shown in Fig. 5. Fig. 5(b), (c), and (d) are the denoised profiles by unbiased preprocessing and 4σ extraction in the spatial domain, $4\sigma_{lf}$ extraction in the frequency domain, and removing the high frequency using threshold, respectively.

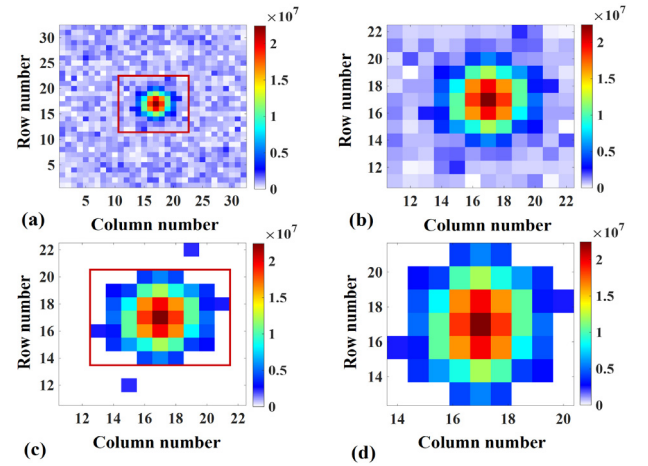


Fig. 4. Illustration of denoising in the frequency domain ($\sigma=6$ mm, SNR=0.2), (a) the amplitude spectrum of the detector signal, (b) $4\sigma_{lf}$ extraction in the frequency domain, (c) removing the high-frequency components by the threshold, (d) discarding the stray high-frequency components.

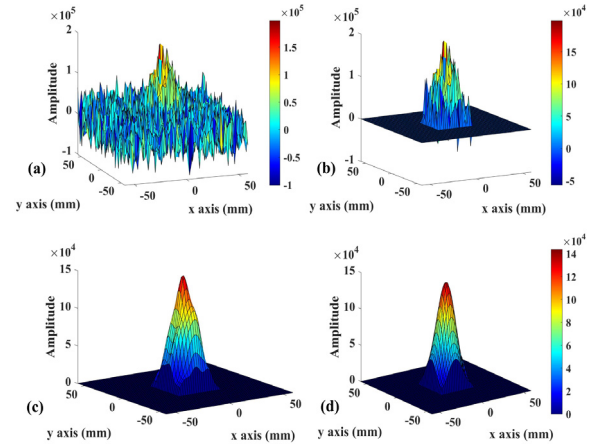


Fig. 5. Example of (a) original beam profile and denoised profiles by (b) unbiased preprocessing and 4σ extraction in the spatial domain, (c) by $4\sigma_{lf}$ extraction in the frequency domain, (d) by removing the high frequency using a threshold. The results were achieved with $\sigma=10$ mm and SNR=0.2.

As shown in Fig. 5(b), the noise on the edge of the beam profile is removed by spatial denoising. Then the burrs are reduced by eliminating the high-frequency components outside $4\sigma_{lf}$ of the frequency spectrum (Fig. 5(c)). Finally, the distortion of the signal is reduced by removing high-frequency components using the threshold (Fig. 5(d)).

4.1. Results of denoising method in the spatial domain

The mean SNR and the mean R^2 for both the original and the denoised signals are presented in Fig. 6 as a function of the beam spot size, changing the σ from 4 to 10 mm. Three examples were considered using original data simulated using three values for SNR (0.2, 0.6, 1.5). The mean SNR is the average value of SNRs for 1000 times of calculations. The mean R^2 is defined similarly.

Overall, the mean SNR and mean R^2 of the spatial denoised signal are higher than those of the original signal. When the beam spot size

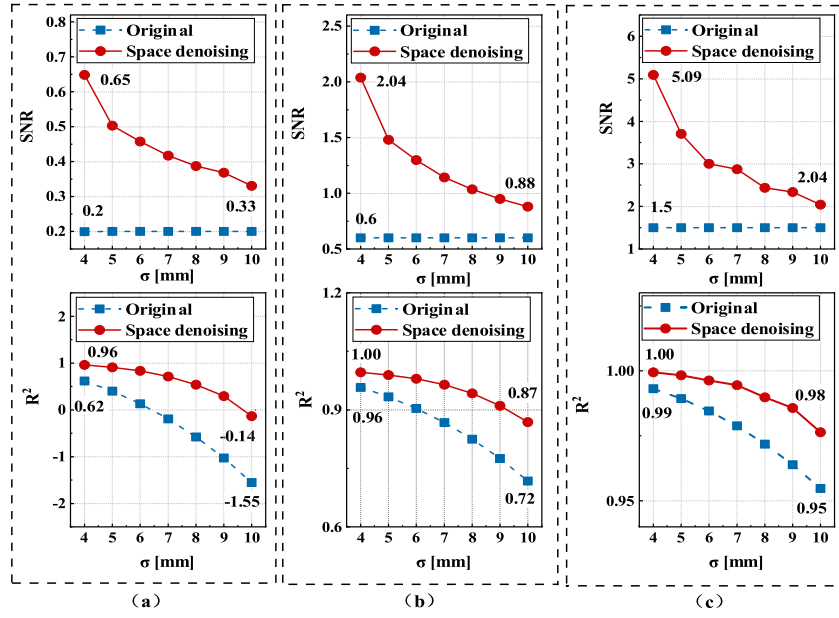


Fig. 6. Behavior of the mean SNR and mean R^2 of the beam distribution for both the original signal and the spatially denoised signal as a function of the beam spot size (σ from 4 to 10 mm) and for different SNRs, (a) 0.2, (b) 0.6, (c) 1.5.

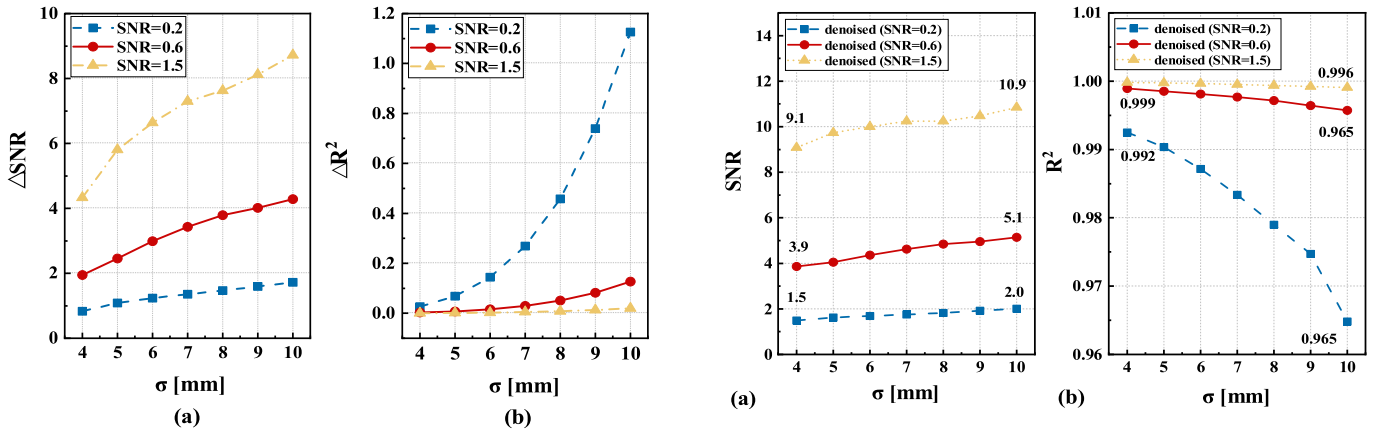


Fig. 7. Behavior of the increment of the mean SNR (ΔSNR) (a) and that of the mean R^2 (ΔR^2) (b) of the beam distribution for the frequency-domain denoised signals as a function of the beam spot size (σ from 4 to 10 mm) and for different SNRs (0.2, 0.6, 1.5).

$\sigma=4$ mm, the SNR of the denoised signal is more than three times larger than that of the original signal and the R^2 of that is higher than 0.96. Besides, the mean SNR and mean R^2 of the spatially denoised signal decrease with the growth of the beam spot size σ .

4.2. Results of denoising method in the frequency domain

The effect of removing the high-frequency components contributed by noise is also discussed. The increment of the mean SNR (ΔSNR) and that of the mean R^2 (ΔR^2) by frequency denoising as a function of the beam spot size σ ($\sigma \in [4, 10]$ mm) under different SNRs ($SNR \in [0.2, 1.5]$) are presented in Fig. 7.

The frequency denoising method works for signals with different SNRs as shown in Fig. 7. The major improvement in term of ΔSNR and ΔR^2 is that the beam profile signal with a larger σ has greater ΔSNR and ΔR^2 by frequency denoising.

4.3. Results of the FFT-based denoising method

The SNR and R^2 using the FFT-based beam profile denoising method under different SNRs as a function of the beam spot size σ ($\sigma \in [4, 10]$ mm) are presented in Fig. 8. Fig. 8 shows that SNRs of the denoised signals are more than six times higher than those of the original signals. R^2 of the denoised signals are larger than 0.96.

4.4. Experimental validation

The FFT-based beam profile denoising method was tested by X-ray radiation experiments. The experimental setup is shown in Fig. 9, including an X-ray source M25 [23] (Newton Scientific, Massachusetts, USA) and a strip ionization chamber IC128. In addition, two multi-channel electrometers I128 [16] comprising a total of 2×128 parallel charge-integrating channels were used to collect signals of 128 channels each of x- and y-directions in the detector.

The SNR of the integrated signal can be improved by prolonging the integration time of the channel currents, which enhances the signal's strength. Therefore, the experiments were carried on under different

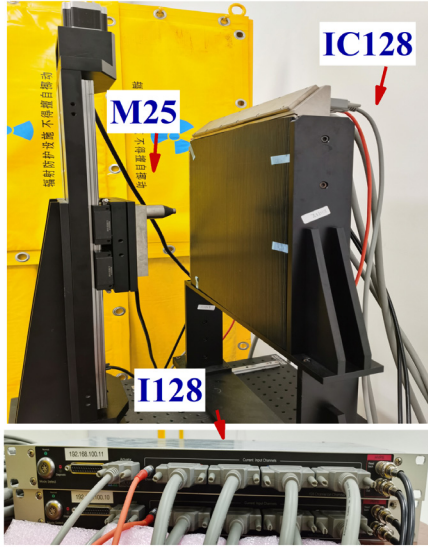


Fig. 9. Experimental instruments, including the X-ray source M25, strip ionization chamber IC128, and two multi-channel electrometers I128.

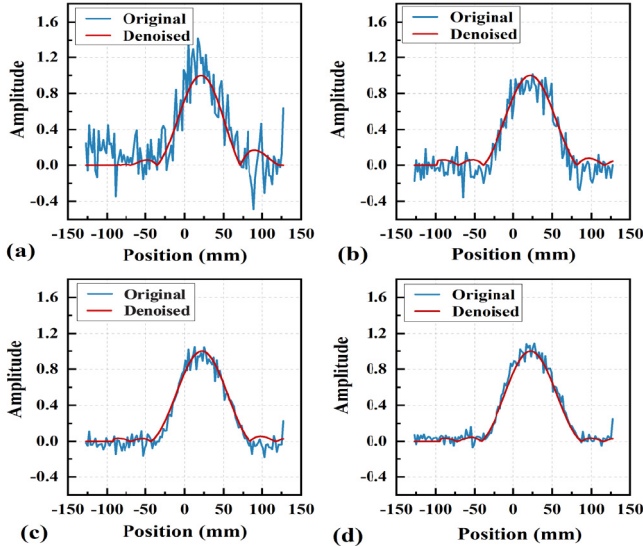


Fig. 10. Both the original and the denoised signals of the detector's 1D projection in the x-direction for different integration times, (a) 100 μ s, (b) 200 μ s, (c) 500 μ s, and (d) 1000 μ s.

integration times: [100, 200, 500, 1000] μ s. The denoised signals of the x-direction under four different integration times are taken as examples and shown in Fig. 10. As shown in Fig. 10, strip signals enhance with the growth of integration time. Thus, the impact of noise on the signals weakens. In addition, the distributions of denoised signals under different integration times have a high degree of consistency.

The mean of the center position $\hat{\mu}$ and spot size $\hat{\sigma}$ obtained by applying Gaussian fitting on the 400 samples in the x- and y-direction, respectively, was considered as the accurate distribution of pencil beam signal shown as red curves in Fig. 11. Each sample that contains data of 128 channels of the multi-strip ionization chamber was obtained when the integration time was 1000 μ s and averaged over 255 samples. As shown in Fig. 11, the curve determined by $\hat{\mu}$ and $\hat{\sigma}$ fits the distribution of the data (the blue bars in Fig. 11) accumulated by strips for 400 samples well.

The performance of the FFT-based beam profile denoising method is evaluated using the criterions R^2 , and the root mean square error

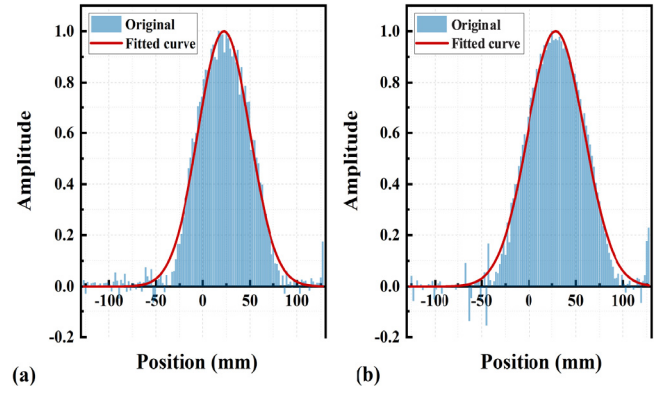


Fig. 11. Real distribution of spot signal in the x- (a) and y- (b) direction. The red curve is determined by the average of the estimated spot center and sigma of 400 samples. The blue bars depict the distribution of the data accumulated by strips for 400 samples. Each sample contains data of 128 channels of the multi-strip ionization chamber.

(RMSE) by comparing the distribution of the denoised signal with the actual distribution of the pencil beam signal. R^2 of the original data and that of the denoised data are presented in Fig. 12(a) as a function of integration time. In addition, RMSEs of μ and σ using the original data and the denoised data as a function of integration time (from 100 to 1000 μ s) are shown in Fig. 12(b) and (c), respectively.

As shown in Fig. 12(a), R^2 of the denoised data is evidently larger than those of the original data, especially when the integration time is 100 μ s. The average of R^2 of the denoised data under different integration times is about 0.96, while that of original data is 0.6. In Fig. 12(b) and (c), when the integration time is 100 μ s, RMSEs of μ and σ using the denoised data are about 1.3 mm and 6.8 mm less than those using the original data.

5. Conclusions

The denoising of detector signals is recommended in particle therapy for accurate measurements of beam profiles mainly when limited beam radiation intensity and short integration time are needed. An FFT-based beam profile denoising method is proposed based on the Gaussian distributions of the pencil beam profile signals in both the spatial and spatial-frequency domains. In this work, we verified the performance of each step of the method able to correct the random bias caused by noise in the spatial domain and deal with the signal fluctuation and distortion in the frequency domain. In conclusion, the presented denoising method improves the accuracy of beam monitors based on segmented ionization chambers and detector arrays without increasing the beam radiation intensity and the channel integration time.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

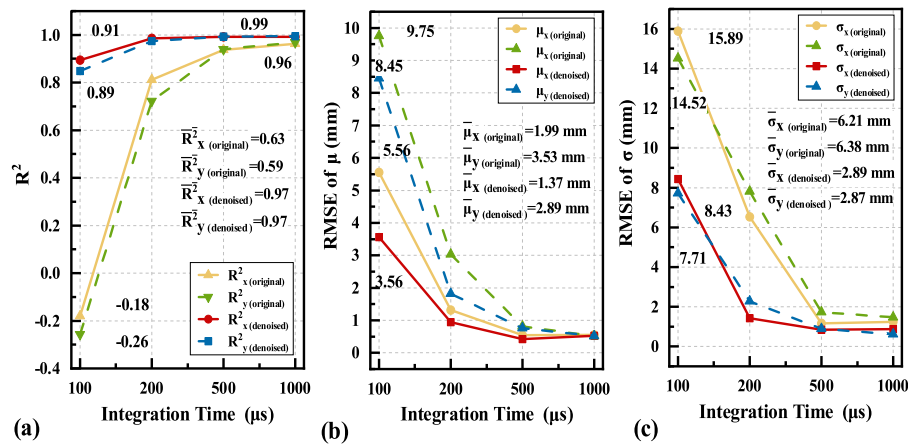


Fig. 12. Behavior of the mean of R^2 (a), RMSEs of μ (b), and RMSEs of σ (c) of the beam distributions for both the original and denoised signals as a function of the integration time (from 100 to 1000 μs).

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